Deep Research

Lunch & Learn
By John Lavender

callibrity

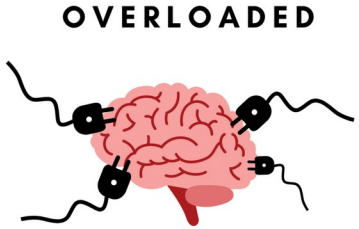
Agenda

1. What is Deep Research?
2. AI-Powered Research: Industry Approaches
3. Foundation Concepts Review
4. Deep Research Implementation
5. GE Aerospace Application
6. Architecture & Technical Deep Dive
7. Future Directions

What is Deep Research?

The Problem Space

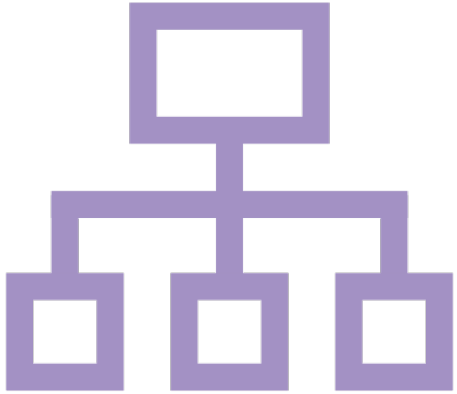
- **Information Overload**: Organizations have vast amounts of unstructured data
- **Context Loss**: Traditional search returns documents, not answers
- **Research Bottleneck**: Manual research is time-consuming and inconsistent
- **Knowledge Silos**: Information trapped across multiple systems



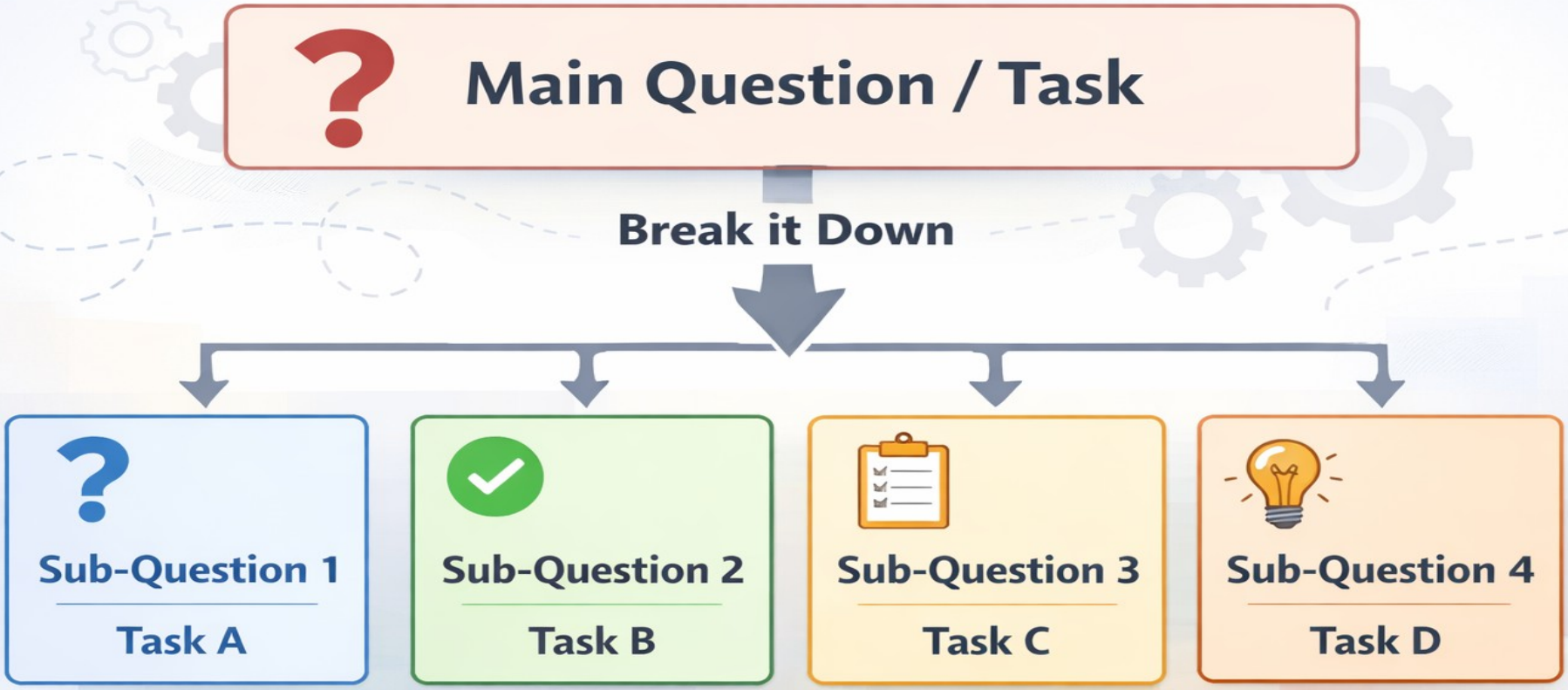
The Solution

Deep Research applies AI to automate complex research tasks by:

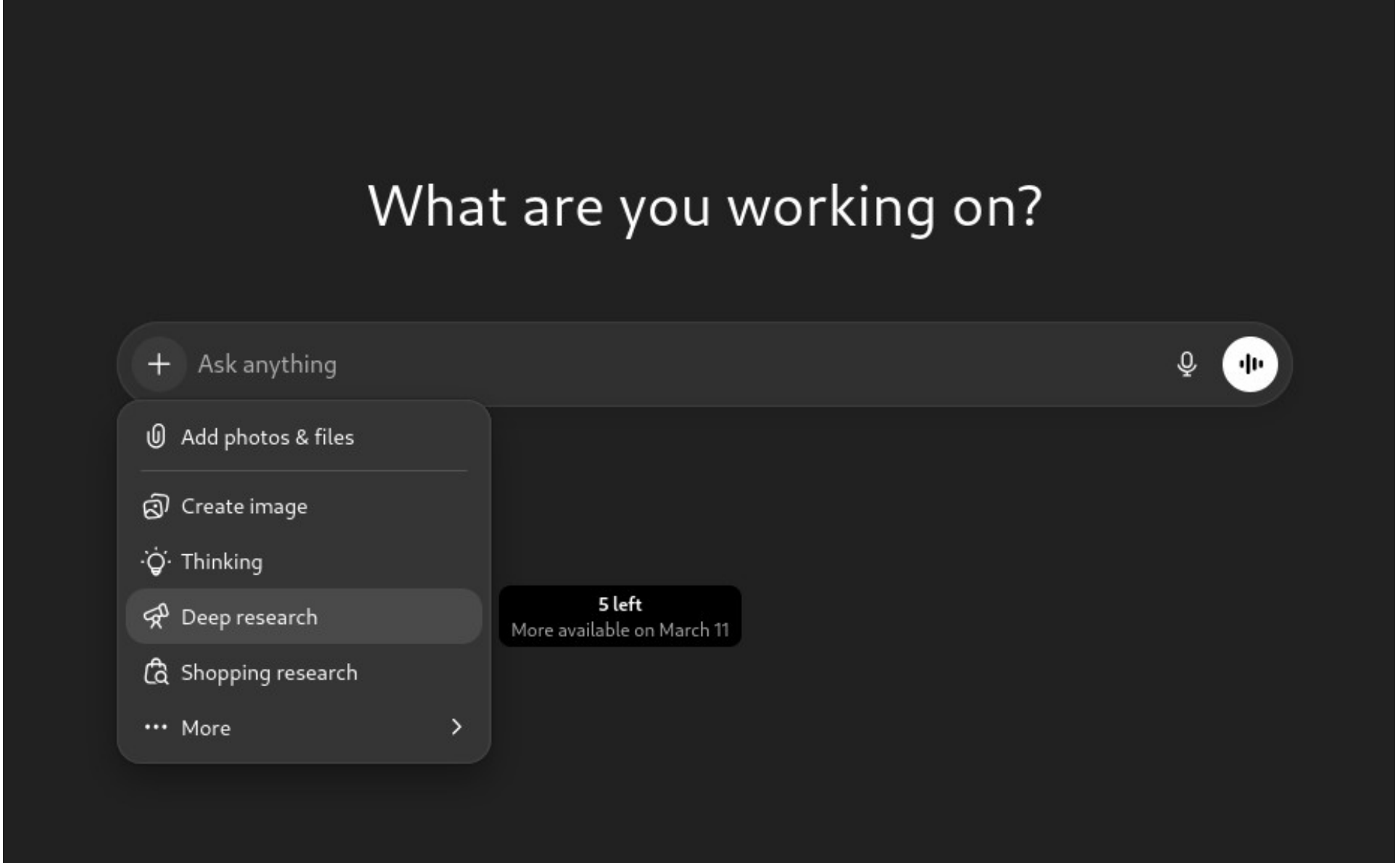
- Breaking down complex queries into manageable subtasks. “Decomposition-based methods achieve strong performance on question-answering tasks” (Radhakrishnan, *Question Decomposition Improves the Faithfulness of Model-Generated Reasoning*).
- Searching relevant knowledge bases intelligently
- Synthesizing findings into coherent insights
- Maintaining context across multi-step reasoning



Task Decomposition



Industry Approach to Deep Research


What are you working on?
Ask anything
Add photos & files
Create image
Thinking
Deep research
Shopping research
More
5 left
More available on March 11

ChatGPT

Memory full Share ··· Activity 29 Sources

I want the 2nd option: a comparison of Marx's original writings with how modern theorists interpret the state today in light of AI and automation?

Great, I'll compare Karl Marx's original conception of the state with modern interpretations by contemporary theorists, especially in the context of AI and automation. This will include how these technologies are reshaping labor, capital, and state power, and how these changes align or diverge from Marx's framework.

I'll share the findings with you shortly. Feel free to ask anything else while I work on this!

Research completed in 7m · 29 sources · 88 searches

2/2

Marx on the State as Class Instrument

Karl Marx described the modern state as inherently bound up with class rule. In the *Communist Manifesto*, he famously wrote that "the executive of the modern state is but a committee for managing the common affairs of the whole bourgeoisie" marxists.org . Elsewhere he defined political power itself as "merely the organised power of one class for oppressing another" marxists.org . In Marx's view, then, the state apparatus (laws, police, bureaucracy) functions to enforce property rights and wage-labour in favor of the capitalist class. Engels and Lenin later echoed this: once classes vanish, the state "loses its political character" marxists.org and eventually "withers away" under communism. Thus Marx saw the state not as neutral but as a vehicle of ruling-class domination, enforcing capitalist interests and thwarting workers' demands.

Technological Change and Production in the AI Era

In Marx's time the means of production were factories and machinery; today they include digital infrastructure – data, algorithms, and advanced hardware. Modern AI systems run on complex microchips and data centers, but as Marxists emphasize, these are still just "dead labour" instruments. For example, a recent analysis notes that although AI systems can perform many tasks, "it is more precise to understand [their] role as intensifying the conditions under which human labour power remains the ultimate source of surplus" nature.com . In other words, machines and algorithms boost productivity but do not create value independently of human labor nature.com . Contemporary theorists extend this by coining terms like *"algorithmic surplus value"* – the extra profits extracted when AI makes human workers more productive – as a 21st-century analogue to Marx's surplus-value nature.com .

The State in the AI Economy: Corporate–State Alliances

In the age of AI, state power often aligns with high-tech capital. Nick Dyer-Witheford and colleagues

Multitasking
Try asking ChatGPT something else while you wait.

Google, Microsoft, IBM, Amazon, Facebook, Alibaba … often "with help thrown in by governments eager for

Ask anything

ChatGPT can make mistakes. Check important info.

OK, let me see. This examines if workers need theory to perceive capitalists' control over state power and how Marx's theory aligns with contemporary state power dynamics.

Searched for Marxist theorists on state capitalism and AI

I'm crafting a Marxist perspective on technology, aligning with ideas from "A Marxist Approach to Technology" by Liberation School.

Read liberationschool.org

It outlines the Marxist analysis of technology and labor in capitalist societies, examining how technological change reshapes labor dynamics.

Read more from liberationschool.org

It seems like there's a critical examination of bourgeois politicians who enlisted state inspectors to study factory conditions. The analysis appears to be informed by Marxist principles.

Read more from liberationschool.org

I'm thinking through Marxist and modern political theorists, considering their insights on state power, labor relations, and the impact of automation and AI on inequality and exploitation.

Searched for AI capitalism and authoritarian state theory

I'm researching connections between Marxist state theory and AI's role in capitalist systems, focusing on labor relations and means of production. Primary and secondary sources will provide insights and perspectives.

Searched for Marx on surveillance state and AI automation

I'm mapping out the focus on the dip and potential trends, considering the implications for future market movements. Progress is steady, with subtle shifts amassing influence.

I'm thinking through state power and digital control. Considering authoritarian and technological shifts, which are surprising. But, digitalization is still reshaping state dynamics.

Combining high-tech hardware and AI, I ponder their roles in advanced production means, embedding an image to illustrate this intersection within at least a three-sentence paragraph.

Read nature.com

Piecing together citations for Marx (18, 45), tech scholars (21, 22, 33, 40, 60), and Asanka (16). Progressing towards finalizing sections and citations.

Research completed in 7m · 29 sources

Capabilities and Limitations

- Key Capabilities:
  - Multi-step reasoning
  - Automatic task decomposition
  - Web search integration
  - Iterative refinement of research
- Limitations:
  - Relies on available web data
  - No enterprise knowledge integration
  - Limited customization for domain-specific needs

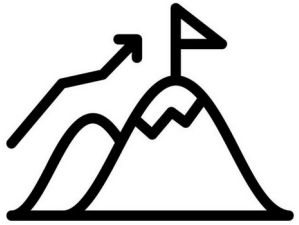
GE Aerospace

Business Context

- Investor Relations team needs to be able to conduct research across a vast database of sensitive internal documents
- Decompose the research tasks to get refined results and to expose the “thought process” to the user

Challenges Addressed

- Domain-specific knowledge requirements
- Configuration of system prompts, parameters, and underlying architecture and dependencies
- Compliance and security considerations
- Scale and performance needs



Challenges Ahead

- Adoption of the product by the IR team
- Training on how to use the product effectively
- Scaling, rate limiting, exponential costs, reliability, and use-case driven tuning
- Integration into the broader GE ecosystem

My Implementation of Deep Research

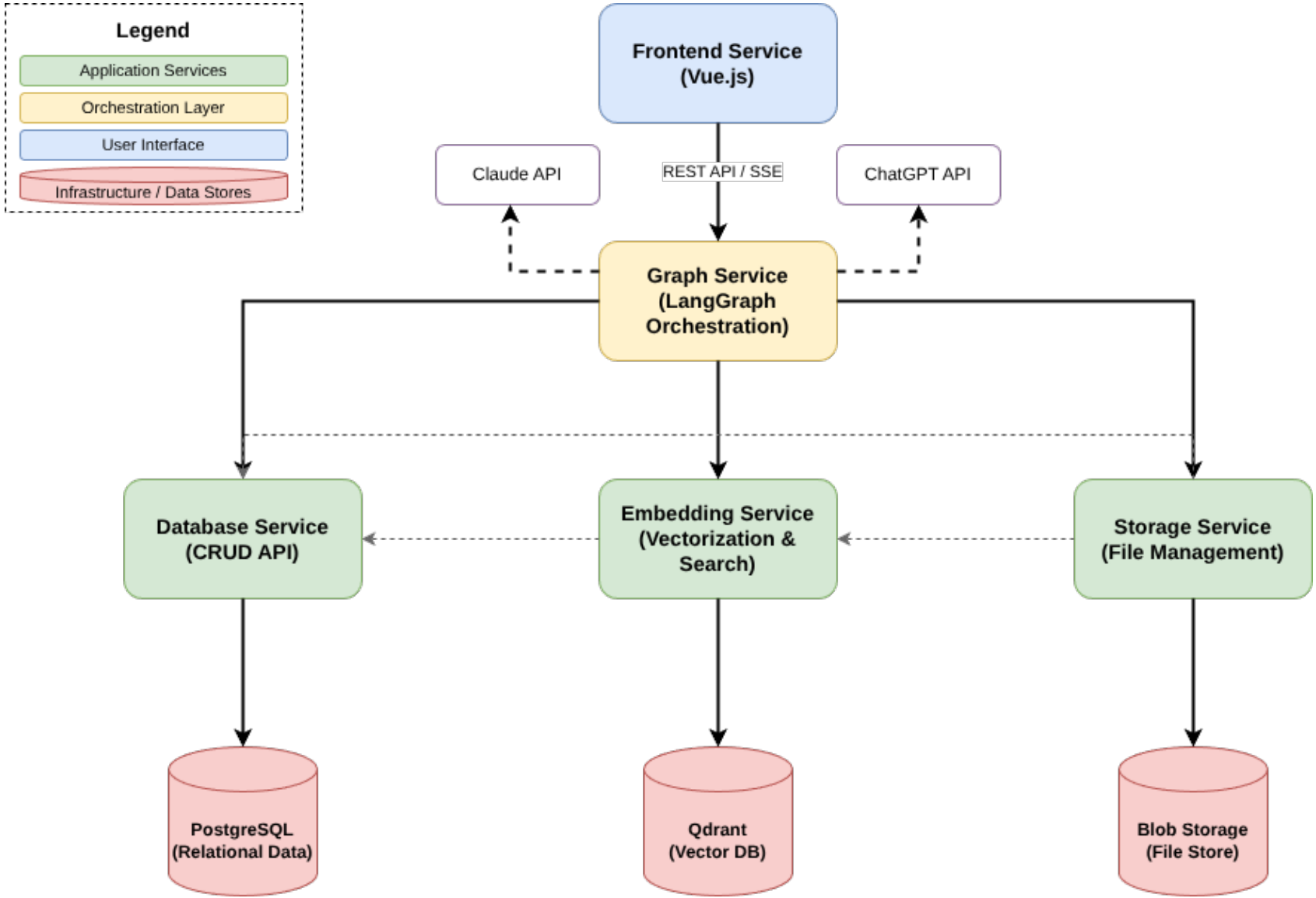
DEEP
RESEARCH

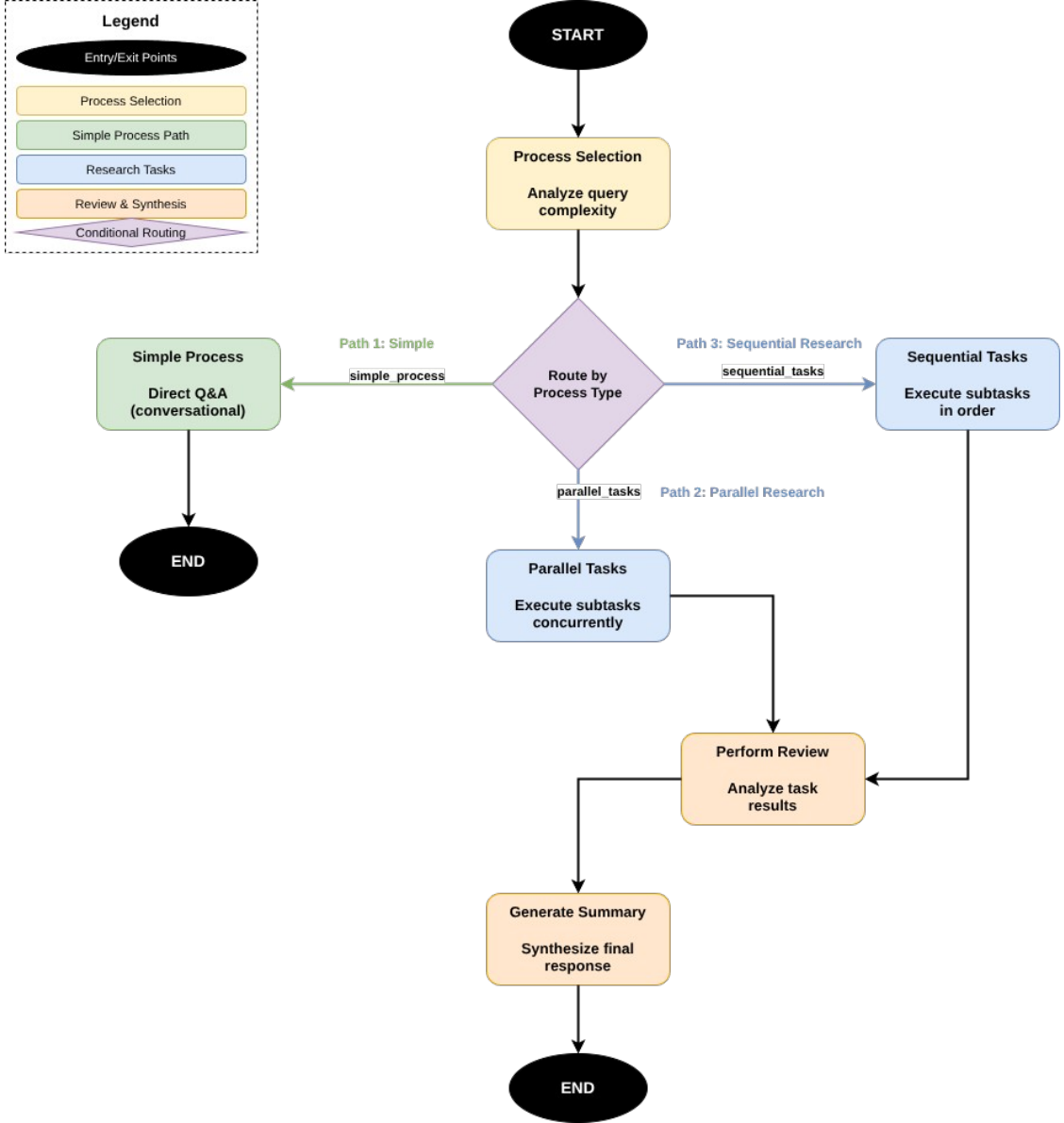
About

- **Independent Implementation**: Original architecture and design — not a ripoff of the GE Aerospace project
- **Local-First Development**: Fully containerized; runs entirely locally with Docker Compose
- **Learning-Focused Scope**: Proof-of-concept demonstrating core capabilities that is not production-ready
- **Modular Architecture**: Microservices design with strong typing and real-time streaming
- **Flexible & Extensible**: Multiple LLM providers; foundation for future enhancements

Obligatory Review of Concepts

- **Chunking**: Breaking documents into smaller segments for efficient storage and retrieval; enables token limit management
- **Vectorization**: Converting text to numerical embeddings that capture semantic meaning; powers similarity search
- **RAG (Retrieval-Augmented Generation)**: Combining vector search with LLM generation; grounds responses in retrieved context to reduce hallucinations
- **LangChain/LangGraph**: Framework for building multi-step AI workflows; provides state management and conditional routing
- **Task Decomposition**: Breaking complex queries into manageable sub-tasks; enables parallel execution and comprehensive research


Legend
Application Services
Orchestration Layer
User Interface
Infrastructure / Data Stores
Frontend Service (Vue.js)
Claude API
REST API / SSE
ChatGPT API
Graph Service (LangGraph Orchestration)
Database Service (CRUD API)
Embedding Service (Vectorization & Search)
Storage Service (File Management)
PostgreSQL (Relational Data)
Qdrant (Vector DB)
Blob Storage (File Store)


Legend
Entry/Exit Points
Process Selection
Simple Process Path
Research Tasks
Review & Synthesis
Conditional Routing
START
Process Selection
Analyze query complexity
Route by Process Type
Path 1: Simple
simple_process
Simple Process
Direct Q&A (conversational)
END
Path 3: Sequential Research
sequential_tasks
Sequential Tasks
Execute subtasks in order
parallel_tasks
Path 2: Parallel Research
Parallel Tasks
Execute subtasks concurrently
Perform Review
Analyze task results
Generate Summary
Synthesize final response
END

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Chat History

Analyze the differences in the definition of 'stat...
Feb 9

Analyze the differences in the definition of 'stat...
Feb 7

+ New Conversation

Analyze the differences in the definition of 'state' between Karl Marx and Max Weber.

02:54 AM

AI Response Completed

- Selected method for answering the query
- Executed 4 research tasks in parallel 4 tasks
- Reviewed research results
- Final summary

Synthesis: what Marx and Weber are *doing* when they "define the state"

The task-results converge on a key meta-finding: Marx and Weber are not offering competing *dictionary* definitions of the same object so much as building two different explanatory projects.

- Marx's "state" talk is mainly diagnostic and critical: the state is intelligible through its *social function* in reproducing a historically specific class society (especially capitalism) and through the *state–civil society split* that makes political power appear as a universal public realm while it stabilizes private property and class domination.
- Weber's "state" talk is mainly analytic and organizational: the state is specified by a distinctive *means* (physical force), a *successful claim to legitimacy*, a *territorial jurisdiction*, and (in modernity) the characteristic *administrative instrument* of rational-legal bureaucracy.

This difference matters because it explains why they often seem to "talk past" one another:

- Marx asks: Whose social order does the state secure, and how is that rooted in political economy?
- Weber asks: What kind of organization is a state, and what stabilizes obedience to it as rule rather than mere violence?

---

1) Marx's conception: state as class power *and* as separated administrative "public order"

Across the Marx-focused results, a coherent conception emerges with three tightly linked layers

Enter your research query...

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Feb 7

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Across the Marx-focused results, a coherent conception emerges with three tightly linked layers.

How are these definitions of 'state' relevant to modern society in the age of AI?

08:59 PM

AI Response Completed

Selected method for answering the query

Executed 5 research tasks in parallel 5 tasks

Analyze how Marx's conception of the state as a class instrument and bureaucratic apparatus applies to AI governance, algorithmic management, and the relationship between tech corporations and state power in modern society

Details

Citations

Examine how Weber's definition of the state as the monopoly of legitimate force and rational-legal bureaucracy relates to AI-driven surveillance, automated decision-making systems, and the changing nature of state administration

Details

Citations

Investigate how the legitimacy and 'imaginary generality' debate between Marx and Weber applies to AI systems that claim neutrality and objectivity while potentially encoding particular interests or biases

Details

Citations

Explore how the question of bureaucratic persistence and the 'withering away' of the state relates to automation, algorithmic governance, and whether AI intensifies or transforms administrative hierarchies

Details

Citations

Enter your research query...

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▼ Details

Marx's State Theory Applied to AI Governance, Algorithmic Management, and Tech-State Power Relations

Core Application Framework

Marx's conception of the state as a class instrument and bureaucratic apparatus provides a powerful analytical lens for understanding contemporary AI governance and the relationship between tech corporations and state power. The key Marxian insights—the state as "committee for managing the common affairs of the whole bourgeoisie," bureaucracy as a "closed society within the state" claiming false universality, and the structural constraints that class power places on state action—translate directly into critical questions about who controls algorithmic systems, whose interests they serve, and how technical expertise functions as a new form of domination.

---

1) The State-Tech Corporation Nexus: A New "Committee" Relationship

Marx's original insight: The state functions as "a committee for managing the common affairs of the whole bourgeoisie"—not as a passive puppet but as an institution that organizes and coordinates ruling class interests while maintaining the appearance of neutrality.

Application to AI governance:

The relationship between tech corporations and state power exhibits precisely this dynamic:

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Karl Mar....pdf p.74 | marx a r....pdf p.21 | Karl Mar....pdf p.199 | marx a r....pdf p.241 | marx a r....pdf p.267
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▼ Reviewed research results

Task Results Review

Overall Assessment

The five tasks collectively provide a comprehensive and intellectually rigorous application of the Marx-Weber state theory comparison to contemporary AI governance issues. The results demonstrate strong analytical depth, effective use of source materials, and meaningful theoretical synthesis.

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1. Completeness Evaluation

Strengths

- All five tasks were successfully completed with substantial analytical content
- Each task addresses its specific focus while maintaining connections to the broader theoretical framework
- The results cover the major dimensions outlined in the original research synthesis: class instrument vs. organizational definition, legitimacy, bureaucracy, and (implicitly) territory

Gaps Identified

- Task 2 (Weber/AI) acknowledges that Weber's canonical inclusion of "territory" is not directly quoted in the source materials—this limitation carries through to Task 5

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Marx and Weber on the State: Implications for AI Governance

Synthesis Overview

The classical debate between Karl Marx and Max Weber on the nature of the state provides a remarkably productive framework for analyzing contemporary AI governance, algorithmic management, and the relationship between technology platforms and state power. This synthesis weaves together findings from five interconnected analyses to reveal how these 19th and early 20th century theoretical frameworks illuminate—and are themselves illuminated by—the challenges of the digital age.

The Core Theoretical Divergence

Marx and Weber were not offering competing dictionary definitions of the same object but building two different explanatory projects:

Marx's approach is diagnostic and critical: The state is intelligible through its *social function* in reproducing class society and through the state–civil society split that makes political power appear universal while stabilizing particular (bourgeois) interests. Bureaucracy operates through an "imaginary generality"—claiming to represent universal interests while actually serving particular class interests.

The "hidden labor" question: If AI systems depend on significant human labor (training, maintenance, gap-filling) that is rendered invisible, how does this complicate both Marxian class analysis and Weberian organizational analysis?

09:16 PM

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09:16 PM

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The Hidden Labor Question: Synthesizing Marx and Weber on AI-Sustaining Work

Executive Summary

This analysis examines how the invisibility of labor sustaining AI systems can be understood through the classical theoretical frameworks of Karl Marx and Max Weber. The central finding is that hidden AI labor creates a structural contradiction: the same mechanisms that enable exploitation (Marx) simultaneously undermine the legitimacy claims upon which AI authority depends (Weber). This contradiction represents both a mechanism of domination and a vulnerability for resistance.

I. The Theoretical Foundations

Marx: The "Hidden Abode" of AI Production

Marx's framework provides three interconnected analytical tools for understanding hidden AI labor:

1. The Labor Theory of Value and Surplus Extraction

Data produced by data annotators represents "a paradigmatic example of commodity production." The extraction mechanism follows classical Marxian lines:

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Limitations and Potential Enhancements

- Graph structure is hard-coded and is not configurable at runtime in the current implementation
- Does not gracefully handle upload of temporary (per chat turn) files
- "Tools" and "skills" are in vogue today - Deep Research can be a "tool" in a larger ecosystem of tools available for agents or humans to invoke

Thank you!

- https://arxiv.org/abs/2307.11768
- https://github.com/JohnLavender474/DeepResearch

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